Software Requirements Specification

for

3AM Talks: AI-Powered Mental Health Assistant

Version: 1.0 Approved

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) provides comprehensive documentation for "3AM Talks: AI-Powered Mental Health Assistant," establishing formal requirements for development, testing, and validation. The document serves as a contract between the development team and supervisors, detailing functional and non-functional requirements necessary for successful implementation.

The SRS demonstrates technical complexity through advanced AI integration including Retrieval-Augmented Generation (RAG), multimodal emotion recognition (facial and vocal), personality detection, 3D avatar rendering, and therapeutic response generation within mental health support contexts.

1.2 Document Conventions

This document follows IEEE Std 830-1998 conventions:

- **Bold text:** Headings and key technical terms
- *Italic text:* Definitions and external references
- **Requirements identifiers:** REQ-FN-XXX (functional), REQ-NF-XXX (non-functional)
- **Priority levels:**
  - **Critical:** Required for core functionality
  - **High:** Essential for core functionality
  - **Medium:** Important but not critical
  - **Low:** Desirable enhancements

1.3 Intended Audience and Reading Suggestions

Primary Audience:

- **Development Team:** Focus on Sections 3, 4, and 6 for implementation details
- **Project Supervisors:** Review Sections 1, 2, and 5 for scope and technical rigor
- **Academic Evaluators:** Complete document review, emphasizing system architecture

Secondary Audience:

- **QA Team:** Sections 4 and 5 for test case derivation
- **System Administrators:** Section 2.4 and deployment models
- **End Users:** Sections 1.4, 2.2, and 2.3 for alignment with user needs

Reading Suggestions:

- **First-time readers:** Begin with Sections 1-2 for context
- **Technical implementers:** Prioritize Sections 3, 4, and 6
- **Project managers:** Focus on Sections 2.5, 2.7, and 5

1.4 Product Scope

Core Problem Addressed:

3AM Talks addresses critical gaps in mental health accessibility within Pakistan and similar contexts:

- **Accessibility:** Limited availability of professionals, especially in rural areas
- **Economic Barriers:** Prohibitive costs for students and young adults
- **Social Stigma:** Cultural barriers preventing help-seeking
- **Temporal Gaps:** Lack of immediate support during crisis moments
- **Cultural Misalignment:** Existing tools lack local cultural sensitivity

Technical Innovation:

The system leverages cutting-edge AI technologies:

- **Specialized LLMs:**
  - PsychoCounsel-Llama3-8B for therapeutic dialogue
  - CBT LLM for cognitive behavioral therapy principles
- **Multimodal Emotion Detection:**
  - MediaPipe/OpenFace for facial expression analysis
  - Hume AI for vocal emotion and prosody analysis
- **Personality Assessment:** BERT-based Big Five model with PsyCoT inference
- **Advanced Speech Processing:**

  - Whisper for accurate real-time transcription
  - Hume EVI-4 for emotion-aware speech synthesis
- **3D Avatar Rendering:** WebGL/Three.js for empathetic visual representation
- **RAG Architecture:** Context-aware responses from CBT, DBT, and ACT literature

Key Objectives:

1. Promote mental health awareness and reduce stigma
2. Provide 24/7 empathetic psychological support
3. Ensure privacy and anonymity
4. Detect crisis situations with appropriate emergency resources
5. Serve as first-line intervention, not professional therapy replacement
6. Demonstrate culturally-sensitive AI in mental health applications

Scope Boundaries:

In Scope:

- Multimodal conversational interface with 3D avatar
- Therapeutic response generation with personality-aware adaptation
- Facial and vocal emotion recognition
- Crisis detection and emergency resources
- Privacy-preserving session management

Out of Scope:

- Medical diagnosis or treatment prescription
- Healthcare provider system integration
- Biometric monitoring beyond facial/vocal analysis
- Real-time human therapist connections
- Electronic health record management

1.5 References

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6. World Health Organization, "Mental Health Atlas 2020," Geneva: WHO Press, 2021.

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2 Overall Description

2.1 Product Perspective

3AM Talks is a self-contained, multi-platform AI mental health assistant integrating external AI services and databases. The system bridges immediate emotional support needs and traditional therapeutic interventions through advanced multimodal AI.

External Dependencies:

- **Specialized LLM Services:** PsychoCounsel-Llama3-8B, CBT LLM
- **Emotion Analysis APIs:** Hume AI for vocal emotion, MediaPipe/OpenFace for facial analysis
- **Speech Processing:** Whisper (ASR), Hume EVI-4 (TTS)
- **Cloud Infrastructure:** AWS/Google Cloud/Azure for scalable deployment
- **Vector Database:** Pinecone/Weaviate for semantic search
- **3D Rendering:** Three.js (r128) for WebGL-based avatar

2.2 Product Functions

2.2.1 Intelligent Conversation Management

- Context-aware dialogue maintaining conversation continuity

- Therapeutic response generation aligned with CBT/DBT/ACT principles
- Multi-turn conversation flow with appropriate intervention timing
- Personality-adaptive communication based on Big Five traits

2.2.2 Multimodal Interaction Capabilities

- Real-time speech recognition with Whisper
- Emotion-aware speech synthesis via Hume EVI-4
- Facial emotion detection through MediaPipe/OpenFace
- Vocal emotion analysis with Hume AI
- 3D avatar rendering using WebGL/Three.js for visual empathy
- Multilingual support (English, with planned Urdu expansion)

2.2.3 Advanced Emotion & Personality Detection

- **Visual Emotion Recognition:** Real-time facial expression analysis detecting seven basic emotions (happiness, sadness, anger, fear, surprise, disgust, neutral)
- **Vocal Emotion Recognition:** Prosody, pitch, and tone analysis for emotional state classification
- **Personality Assessment:** BERT-based Big Five trait detection (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism)
- **Multimodal Fusion:** Integration of visual, vocal, and textual cues for comprehensive emotional understanding

2.2.4 3D Avatar Representation

- WebGL-based real-time rendering
- Emotionally responsive facial animations synchronized with speech
- Customizable avatar appearance for user comfort
- Performance-optimized for mobile devices

2.2.5 Knowledge-Based Therapeutic Guidance

- RAG-powered responses from evidence-based literature
- Personalized interventions based on personality and emotional state

- Psychoeducational content delivery

2.2.6 Safety and Crisis Management

- Automated crisis detection (self-harm, suicide ideation)
- Emergency resource provision
- Professional referral guidance

2.2.7 User Experience and Data Management

- Encrypted session history
- Optional mood tracking with visualizations
- Privacy-preserving analytics

2.3 User Classes and Characteristics

Primary User Classes:

Class 1: Students and Young Adults (16-30)

- **Demographics:** University students, early career professionals
- **Technical Proficiency:** High comfort with digital platforms
- **Primary Concerns:** Academic stress, career anxiety, relationships
- **Usage Patterns:** Frequent short sessions, late-night usage, crisis intervention during exams

Class 2: Working Professionals (25-45)

- **Demographics:** Mid-career professionals, parents
- **Primary Concerns:** Work-life balance, burnout, family stress
- **Usage Patterns:** Time-efficient sessions, practical coping strategies

Class 3: Rural/Semi-Urban Residents

- **Demographics:** Limited mental health infrastructure access
- **Technical Proficiency:** Variable, requires intuitive interfaces
- **Primary Concerns:** Isolation, economic stress, traditional value conflicts
- **Usage Patterns:** Infrequent intensive sessions, gradual trust building

Secondary User Classes:

- **System Administrators:** Technical maintenance and monitoring
- **Mental Health Professionals:** Content validation and crisis protocol development
- **Researchers:** Academic evaluation and outcome analysis

2.4 Operating Environment

Client-Side Environment:

Platform	Minimum Version	Specifications	Requirements
Android	8.0 (API 26)	4GB RAM, 32GB storage	Microphone, camera, internet
iOS	13.0	iPhone 7+, 3GB RAM	Speech framework, camera access
Web	Chrome 80+, Firefox 75+, Safari 13+, Edge 80+	WebGL 2.0, WebRTC	Modern GPU for 3D rendering

Table 1: Client-Side Operating Environment

Server-Side Environment:

- **Primary Platform:** AWS (multi-region deployment)
- **Backend:** Python 3.9+, FastAPI
- **AI/ML Stack:**
  - Transformers (HuggingFace) for LLM integration
  - LangChain for RAG architecture
  - PyTorch for custom models
  - MediaPipe/OpenFace for facial analysis
  - Librosa for audio processing
  - Three.js (r128) for 3D avatar rendering
- **Databases:**
  - PostgreSQL for structured data
  - Vector Database (Pinecone/Weaviate) for semantic search

- Redis for caching

Network Requirements:

- **Minimum Bandwidth:** 256 kbps (text), 1 Mbps (voice + video)
- **Latency:** <200ms for real-time interaction
- **Connectivity:** 4G/5G, Wi-Fi

2.5 Design and Implementation Constraints

Technical Constraints:

- **Response Time:** End-to-end latency <3 seconds
- **Concurrent Users:** Minimum 1,000 without performance degradation
- **Mobile Optimization:** App size <150 MB, battery-efficient
- **3D Rendering:** Target 30 FPS on mobile devices for avatar animations
- **Bandwidth Limitations:** Function on 2G/3G networks (rural areas)

Regulatory Constraints:

- **Non-Clinical Limitation:** No medical diagnosis or treatment recommendations
- **Crisis Protocols:** Mandatory emergency response procedures
- **Data Protection:** GDPR compliance, Pakistani data protection laws

Cultural & Ethical Constraints:

- **Religious Sensitivity:** Respectful of Islamic values
- **Cultural Appropriateness:** Family-oriented, collectivist considerations
- **Transparency:** Users informed of AI interaction
- **Bias Prevention:** Training data reviewed for demographic biases

Resource Constraints:

- **Timeline:** 9-month academic completion
- **Team:** Four-person development team
- **Budget:** Minimal for external services

2.6 User Documentation

End-User Documentation:

- Interactive tutorial demonstrating multimodal features including avatar interaction
- Quick start guide with visual instructions
- Safety disclaimer and crisis resources directory
- Feature-specific guides (voice, camera permissions, avatar customization)

Technical Documentation:

- Deployment manual with 3D rendering optimization
- API documentation for all system interfaces
- Integration guidelines for external AI services
- Testing procedures including emotion recognition validation

2.7 Assumptions and Dependencies

User Assumptions:

- Basic smartphone operation skills
- Comfort with camera and microphone permissions for multimodal interaction
- Willingness to engage in structured therapeutic conversations
- Acceptance of anonymous data collection with transparency

Technical Dependencies:

- **LLM Services:** PsychoCounsel-Llama3-8B, CBT LLM availability (99%+ uptime)
- **Emotion Analysis:** Hume AI, MediaPipe/OpenFace service reliability
- **Speech Processing:** Whisper, Hume EVI-4 accuracy standards
- **Cloud Infrastructure:** AWS/GCP/Azure service levels
- **WebGL Support:** Modern browser/device GPU capabilities for 3D rendering

Organizational Dependencies:

- Faculty guidance throughout project lifecycle
- Access to computing resources

- Mental health professional consultation
- Ethics approval for sensitive conversations

3 External Interface Requirements

3.1 User Interfaces

3.1.1 Mobile Application Interface

Home Screen Requirements:

UI-REQ-001: Home screen displays prominent "Start Conversation" button (30% screen area) with calming color palette (WCAG 2.1 AA compliant: #87CEEB, #90EE90, contrast ratio $\geq$4.5:1). *Priority: High*

UI-REQ-002: Emergency resources accessible within two taps from any screen. *Priority: Critical*

Conversation Interface Requirements:

UI-REQ-003: 3D avatar displayed prominently with real-time emotional expressions synchronized to AI responses. Avatar facial animations driven by emotion detection results. *Priority: High*

UI-REQ-004: Real-time status indicators: "Listening," "Analyzing Emotion," "Processing," "Speaking" with visual waveform and facial expression confidence scores. *Priority: High*

UI-REQ-005: Optional camera view for facial emotion detection with privacy toggle. Clear visual feedback when camera is active. *Priority: High*

UI-REQ-006: Pause button for temporary session stop without ending conversation. *Priority: Medium*

Dashboard Interface:

UI-REQ-007: Mood tracking visualizations (line graphs, emoji indicators) with personality trait insights from Big Five analysis. *Priority: Medium*

UI-REQ-008: Session history with timestamps, duration, dominant emotions, and delete options. *Priority: Medium*

Crisis Alert Interface:

UI-REQ-009: Full-screen crisis modal (high-contrast red #DC143C on white) with emergency contacts and one-tap calling functionality. *Priority: Critical*

3.1.2 Web Application Interface

UI-REQ-010: Responsive design supporting desktop (1920x1080), tablet (768x1024), mobile (375x667). WebGL-enabled 3D avatar rendering across all platforms. *Priority: High*

UI-REQ-011: Clear microphone and camera permission requests with explanatory text. *Priority: High*

3.1.3 Accessibility Requirements

UI-REQ-012: Keyboard-navigable interactive elements with screen reader support. *Priority: High*

UI-REQ-013: Adjustable text size (16px-24px) maintaining layout integrity. *Priority: Medium*

3.2 Hardware Interfaces

3.2.1 Microphone Interface

HW-REQ-001: Interface with device microphones via standard OS APIs (AVFoundation iOS, MediaRecorder Web, AudioRecord Android).

- Sample Rate: 16 kHz minimum (44.1 kHz recommended)
- Bit Depth: 16-bit PCM
- Format: WAV/AAC/Opus

Priority: High

HW-REQ-002: Implement noise cancellation and automatic gain control. *Priority: Medium*

3.2.2 Camera Interface

HW-REQ-003: Interface with device front-facing camera for facial emotion detection.

- Resolution: 640x480 minimum (720p recommended)
- Frame Rate: 15 FPS minimum (30 FPS optimal)
- Format: MJPEG/H.264

Priority: High

HW-REQ-004: Privacy controls allowing users to disable camera while maintaining voice interaction. *Priority: High*

3.2.3 Audio Output Interface

HW-REQ-005: Output synthesized speech through speakers/headphones using standard OS APIs.

- Sample Rate: 22.05 kHz minimum
- Format: MP3/AAC
- Latency: <100ms

Priority: High

3.2.4 GPU Interface

HW-REQ-006: Access device GPU for WebGL-based 3D avatar rendering with automatic quality adjustment based on device capabilities. *Priority: High*

3.3 Software Interfaces

3.3.1 Specialized LLM Interfaces

SW-REQ-001: Interface with PsychoCounsel-Llama3-8B for therapeutic dialogue generation via REST API (HTTPS, Bearer Token auth, JSON format, 30s timeout). *Priority: Critical*

SW-REQ-002: Interface with CBT LLM for structured cognitive behavioral therapy interventions. *Priority: Critical*

SW-REQ-003: Implement exponential backoff retry (max 3 attempts) for failed API requests. *Priority: High*

3.3.2 Emotion Recognition Interfaces

SW-REQ-004: Integrate MediaPipe/OpenFace for real-time facial emotion detection.

- Input: Video frames (640x480, 15+ FPS)
- Output: Emotion probabilities for 7 classes + facial landmarks
- Latency: <100ms per frame

Priority: Critical

SW-REQ-005: Integrate Hume AI for vocal emotion analysis.

- Input: Audio segments (16 kHz, mono)
- Output: Emotion classifications with prosody features

- Latency: <200ms

Priority: Critical

SW-REQ-006: Implement multimodal emotion fusion combining facial, vocal, and textual cues with weighted averaging based on confidence scores. *Priority: High*

3.3.3 Personality Detection Interface

SW-REQ-007: Integrate BERT-based Big Five personality model with PsyCoT inference.

- Input: Conversation history (minimum 5 turns)
- Output: Five trait scores (0-1 scale) with confidence intervals
- Update frequency: Every 10 conversation turns

Priority: Medium

3.3.4 Speech Processing Interfaces

SW-REQ-008: Integrate Whisper for ASR with <10% WER for clear speech, <2s latency for 30s utterances. *Priority: Critical*

SW-REQ-009: Integrate Hume EVI-4 for emotion-aware TTS.

- Input: Text + emotion parameters (valence, arousal, dominance)
- Output: Expressive speech (MP3, streaming)
- Latency: <500ms for first chunk
- MOS: >4.0 for naturalness

Priority: Critical

3.3.5 3D Avatar Rendering Interface

SW-REQ-010: Implement Three.js (r128) WebGL-based 3D avatar system.

- Rendering: 30 FPS target on mobile, 60 FPS on desktop
- Avatar models: Low-poly optimized (<10k triangles)
- Facial animations: Blend shapes for 7 basic emotions
- Lip-sync: Synchronized with TTS audio output

Priority: High

SW-REQ-011: Implement automatic quality scaling based on device performance metrics (FPS, GPU load). *Priority: Medium*

3.3.6 Vector Database Interface

SW-REQ-012: Interface with vector database (Pinecone/Weaviate/FAISS) for semantic search.

- Query: 768-dimensional embeddings (BERT-based)
- Response: Top-k documents (k=3-5) with metadata
- Similarity: Cosine similarity
- Latency: <100ms

Priority: Critical

3.3.7 Database Interfaces

SW-REQ-013: PostgreSQL interface for structured data (SSL/TLS, SQLAlchemy ORM, connection pooling, 5s query timeout). *Priority: High*

SW-REQ-014: Redis interface for caching and session state management. *Priority: Medium*

3.4 Communications Interfaces

3.4.1 Network Protocols

COM-REQ-001: All client-server communication via HTTPS (TLS 1.2 minimum, TLS 1.3 recommended). *Priority: Critical* **COM-REQ-002:** WebSocket Secure (WSS) for real-time bidirectional streaming (voice, video, emotion data).

- Frame Size: 4096 bytes max
- Keep-Alive: Ping/Pong every 30s
- Auto-reconnection with exponential backoff

Priority: High

3.4.2 Data Transmission Formats

COM-REQ-003: JSON for structured data (API requests/responses, metadata, emotion classifications). *Priority: High* **COM-REQ-004:** Binary/Base64 for audio data

with MIME type headers (audio/wav, audio/mpeg, audio/ogg). *Priority: High* **COM-REQ-005:** Protocol Buffers for video frame transmission (efficient serialization for facial emotion detection). *Priority: Medium*

3.4.3 Core API Endpoints

COM-REQ-006: RESTful API following standard HTTP methods:

- POST /api/v1/sessions - Create conversation session
- POST /api/v1/sessions/id/messages - Send message
- POST /api/v1/audio/transcribe - Whisper transcription
- POST /api/v1/audio/synthesize - Hume EVI-4 synthesis
- POST /api/v1/emotion/facial - MediaPipe emotion detection
- POST /api/v1/emotion/vocal - Hume AI emotion analysis
- GET /api/v1/personality/session$_id - Retrievepersonalitytraits$
- GET /api/v1/sessions/id/history - Conversation history
- DELETE /api/v1/sessions/id - Delete session
- GET /api/v1/resources/emergency - Emergency contacts

Priority: High

4 System Features

4.1 Real-Time Voice Interaction (ASR & TTS)

4.1.1 Description and Priority

Priority: **High**

Real-time voice interaction enables natural conversational engagement through bidirectional audio communication: speech recognition (*Whisper*) and emotion-aware synthesis (*Hume EVI-4*). This feature is critical for accessibility and emotional expressiveness.

4.1.2 Stimulus/Response Sequences

Stimulus	**System Response**	**Timing**
User presses "Start Talking"	Microphone activates, "Listening" indicator	<200ms
User speaks (3-30s)	Audio capture, waveform visualization	Real-time
User stops speaking (VAD)	Finalize capture, "Processing" indicator, send to Whisper	<100ms
Whisper returns transcription	Display text, process through RAG + emotion detection	<500ms
AI response generated	Hume EVI-4 TTS with emotional prosody, avatar lip-sync	<1000ms
User taps "Interrupt"	Stop playback, return to listening	<50ms

4.1.3 Functional Requirements

- **REQ-FN-001**: Capture audio at ≥16 kHz, 16-bit PCM. *Priority: High*
- **REQ-FN-002**: Whisper ASR achieving WER <10% (clear speech), <15% (conversational with noise), <2s latency. *Priority: High*
- **REQ-FN-003**: Voice Activity Detection (VAD) identifies speech endpoints within 300ms, false positive rate <5%. *Priority: High*
- **REQ-FN-004**: Hume EVI-4 TTS with MOS >4.0, <1s latency for first chunk, emotional tone modulation (calm, encouraging, validating). *Priority: High*
- **REQ-FN-005**: Real-time interruption ("barge-in"): audio stops within 50ms, microphone activates within 100ms. *Priority: High*

- **REQ-FN-006**: Visual feedback during states (listening, processing, speaking) with animations updated at 30 FPS. *Priority: Medium*
- **REQ-FN-007**: Noise suppression and echo cancellation (10-15 dB reduction). *Priority: Medium*

4.2 Therapeutic Response Generation (RAG)

4.2.1 Description and Priority

Priority: **High**

RAG architecture ensures responses grounded in evidence-based therapeutic frameworks (CBT, DBT, ACT) by retrieving relevant literature before generating responses with PsychoCounsel-Llama3-8B and CBT LLM.

4.2.2 Stimulus/Response Sequences

Stimulus	**System Response**	**Timing**
User utterance transcribed	Extract semantic content, generate query embedding	<100ms
Query embedding generated	Vector DB retrieves top-k (3-5) therapeutic documents	<100ms
Documents retrieved	Construct augmented prompt (user input + context + history + personality traits)	<50ms
Prompt ready	PsychoCounsel-Llama3-8B/CBT LLM generates response	1-2s
Response generated	Safety checks, content filtering, convert to speech	<500ms

4.2.3 Functional Requirements

- **REQ-FN-008**: Knowledge base of ≥500 vectorized therapeutic documents (CBT, DBT, ACT) with 200-500 token chunks, embedded using consistent model (all-MiniLM-L6-v2). *Priority: High*
- **REQ-FN-009**: Semantic search retrieves top-k (k=3-5) documents with cosine similarity >0.7, <100ms latency (95th percentile), >85% retrieval accuracy. *Priority: High*
- **REQ-FN-010**: Construct prompts including: (1) system instructions defining therapeutic persona, (2) retrieved context, (3) last 5 conversation turns, (4) personality trait summary, (5) current user input. Total prompt <4000 tokens. *Priority: High*

- **REQ-FN-011**: LLMs (PsychoCounsel-Llama3-8B, CBT LLM) generate empathetic, non-judgmental responses aligned with therapeutic principles. Empathy score >4.0 (mental health professional validation), zero medical diagnosis, 50-200 words. *Priority: High*
- **REQ-FN-012**: Content filtering detects harmful responses with >95% accuracy, <50ms latency. Filtered responses replaced with safe fallback. *Priority: Critical*
- **REQ-FN-013**: Knowledge base updates via admin interface supporting PDF/DOCX/TXT with automated embedding generation. *Priority: Medium*
- **REQ-FN-014**: Log retrieval operations (query, documents, similarity scores, response) with encryption and pseudonymization. *Priority: Medium*

4.3 Emotion & Intent Detection

4.3.1 Description and Priority

Priority: **High**

Multimodal emotion detection analyzes facial expressions (*MediaPipe/OpenFace*), vocal characteristics (*Hume AI*), and textual content to understand emotional state. Personality detection (*BERT Big Five*) enables adaptive communication. Avatar expressions synchronized with detected emotions.

4.3.2 Stimulus/Response Sequences

Stimulus	System Response	Timing
User speaks with emotional characteristics	Hume AI analyzes prosody, pitch, tone; classifies emotion	<200ms
Camera captures facial expressions	MediaPipe/OpenFace detects facial landmarks, classifies emotion	<100ms
ASR transcription available	Text-based emotion classifier analyzes semantics	<100ms
Multi-modal predictions	Fusion of facial, vocal, textual cues; determine dominant emotion	<50ms
Emotion classified (e.g., "Sadness: 0.85")	Adapt LLM prompt, update 3D avatar expression, adjust TTS prosody	<10ms
Intent detected	Prioritize retrieval strategy (coping for help-seeking, validation for venting)	<50ms
Personality traits updated	Adjust communication style (formal for low openness, casual for high extraversion)	Asynchronous

4.3.3 Functional Requirements

- **REQ-FN-015**: Classify user emotional state into 7 categories (Neutral, Happy, Sad, Angry, Anxious, Fearful, Surprised) with confidence scores. Accuracy >70% on standard datasets (IEMOCAP, AffectNet). *Priority: High*
- **REQ-FN-016**: Facial Emotion Recognition via *MediaPipe/OpenFace*:
  - Input: Video frames (640x480, 15+ FPS)
  - Output: Emotion probabilities + facial landmarks
  - Latency: <100ms per frame
  - Accuracy: >75% on FER2013 dataset

  Priority: High
- **REQ-FN-017**: Vocal Emotion Recognition via *Hume AI*:
  - Input: Audio segments (16 kHz, mono)
  - Features: F0, MFCCs, spectral centroid, energy, prosody
  - Output: Emotion classification with arousal/valence dimensions
  - Latency: <200ms
  - Accuracy: >70% on EmoV-DB

  Priority: High
- **REQ-FN-018**: Textual Emotion Analysis using NLP:
  - Sentiment analysis accuracy >75%
  - Distress keyword detection >90% recall
  - Processing latency <100ms

  Priority: Medium
- **REQ-FN-019**: Multimodal Emotion Fusion: Weighted averaging of facial (40%), vocal (40%), textual (20%) predictions. Weights adjustable based on confidence scores. *Priority: High*
- **REQ-FN-020**: Personality Detection using BERT-based Big Five model:
  - Input: Minimum 5 conversation turns
  - Output: Scores for Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism (0-1 scale with confidence intervals)

  - Update frequency: Every 10 turns
  - Accuracy: >65% correlation with standardized personality assessments

  Priority: Medium

- **REQ-FN-021**: 3D Avatar Expression Synchronization:
  - Map detected emotions to avatar blend shapes
  - Smooth transitions between expressions (300ms interpolation)
  - Maintain 30 FPS rendering during expression changes

  Priority: High

- **REQ-FN-022**: Tag each conversation turn with: dominant emotion, confidence, secondary emotions, personality trait snapshot. Store with timestamp and session ID. *Priority: Medium*

- **REQ-FN-023**: Dynamically adjust LLM prompts based on detected emotion:
  - Sadness → validation-focused prompts
  - Anger → de-escalation prompts
  - Anxiety → grounding techniques
  - Adjust based on personality (high neuroticism → extra reassurance)

  Priority: High

- **REQ-FN-024**: Detect user intent: seeking advice, venting, requesting information, expressing crisis, small talk. Classification accuracy >75%. *Priority: Medium*

- **REQ-FN-025**: Emotion tracking visualization with personality insights dashboard (line graphs, heatmaps, Big Five radar charts). *Priority: Low*

4.4 User Dashboard & Session History

4.4.1 Description and Priority

Priority: **Medium**

Dashboard provides personalized interface for reviewing conversations, tracking emotional patterns, personality trait evolution, and managing data privacy.

4.4.2 Stimulus/Response Sequences

Stimulus	System Response	Timing
User navigates to "History"	Retrieve session list (date, duration, emotions, personality snapshot)	<500ms
User selects session	Display transcript with timestamps, emotion tags, avatar expression history	<300ms
User requests deletion	Confirmation dialog → permanent deletion of all data	<100ms
User accesses analytics	Generate emotion trends, personality evolution, engagement graphs	<1s

4.4.3 Functional Requirements

- **REQ-FN-026**: Securely store conversation logs locally (device encryption) or cloud (AES-256), user-selectable during onboarding. *Priority: Medium*
- **REQ-FN-027**: Permanent deletion capability (individual sessions or all history) with confirmation, GDPR-compliant. *Priority: High*
- **REQ-FN-028**: Session history chronologically ordered with date, time, duration, dominant emotion, personality traits. Supports pagination, search by date/emotion. *Priority: Medium*
- **REQ-FN-029**: User-defined notes/tags (up to 500 characters) with same security as conversation data. *Priority: Low*
- **REQ-FN-030**: Summary analytics: total sessions, average duration, emotion distribution (pie/bar charts), engagement trends (weekly/monthly), personality trait evolution (radar charts comparing early vs. recent sessions). *Priority: Low*

4.5 Crisis Detection & Safety Redirection

4.5.1 Description and Priority

Priority: **Critical**

Crisis detection identifies suicidal ideation, self-harm intent, or imminent danger through keyword analysis, intent classification, and severity scoring. Immediate safety protocols activated with emergency resources.

4.5.2 Stimulus/Response Sequences

Stimulus	System Response	Timing
Crisis keywords ("suicide," "end it all")	Flag utterance as high-risk	<50ms
Crisis flag triggered	Bypass RAG, load pre-scripted crisis response	<10ms
Crisis response loaded	Full-screen modal with emergency contacts, validation message, 3D avatar displays concerned expression	<100ms
Crisis modal displayed	One-tap calling for crisis hotlines, emergency services	Immediate
Crisis event logged	Record event (timestamp, context, risk level) - anonymized	Async
User acknowledges modal	Option to continue (heightened monitoring) or exit with resources	User-paced

4.5.3 Functional Requirements

- **REQ-FN-031**: Analyze every input for high-risk keywords/phrases ($\geq$100 validated crisis indicators). Recall >95%, precision >80%, latency <50ms. *Priority: Critical*
- **REQ-FN-032**: Multi-mechanism detection: keyword matching, intent classification (ML model trained on crisis datasets), severity scoring (combines signals $\rightarrow$ low/medium/high risk). *Priority: Critical*
- **REQ-FN-033**: Crisis protocol activation within 100ms, bypassing normal pipeline. Pre-scripted responses reviewed by mental health professionals. *Priority: Critical*
- **REQ-FN-034**: Full-screen crisis modal (high-contrast: red #DC143C header on white, font $\geq$18pt) with region-specific emergency contacts (geolocation-based). *Priority: Critical*

- **REQ-FN-035**: One-tap calling functionality (`tel:` protocol) for crisis hotlines and emergency services. *Priority: Critical*
- **REQ-FN-036**: Compassionate crisis messaging: "You're not alone," "Help is available," "This is temporary," clear call-to-action. *Priority: Critical*
- **REQ-FN-037**: Log crisis events (`event_id`, `timestamp`, `risk_level`, `trigger_keywords`, `user_pseudonym`) with encryption, accessible only to authorized admins. Regular review process. *Priority: High*
- **REQ-FN-038**: Post-acknowledgment options: continue with heightened monitoring (crisis resource reminder in responses) or exit with comprehensive resource list. *Priority: Medium*
- **REQ-FN-039**: Regularly updated crisis resource database (hotlines, emergency services, local facilities) for Pakistan, India, USA, UK (minimum). Quarterly verification. *Priority: Medium*

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- **REQ-NF-001**: End-to-end response time from user finishing speech to AI speaking ≤3 seconds (>1 Mbps, <100ms latency). Breakdown: VAD <300ms, ASR <1000ms, Emotion + Retrieval + LLM <1500ms, TTS <500ms, Avatar Animation <200ms. *Priority: High*
- **REQ-NF-002**: Support ≥1,000 concurrent users without >10% performance degradation. 95th percentile response time <3.3s, error rate <1%. Horizontal scaling. *Priority: High*
- **REQ-NF-003**: RAG retrieval relevance ≥90% (expert evaluation, average score >4.0 on 1-5 scale, <10% queries scoring <3). *Priority: High*
- **REQ-NF-004**: 99% uptime monthly (max 7.2 hours unplanned downtime). Scheduled maintenance <4 hours/month with 72-hour notice. *Priority: Medium*
- **REQ-NF-005**: Graceful degradation when external services unavailable: detect within 5s, display informative error with emergency contacts, provide offline resource access, auto-retry with exponential backoff. *Priority: Medium*
- **REQ-NF-006**: Mobile app size ≤150 MB (Android APK, iOS IPA). On-demand downloading for large resources. *Priority: Medium*
- **REQ-NF-007**: 3D avatar rendering: 30 FPS minimum on mobile (720p), 60 FPS on desktop (1080p). Automatic quality scaling based on device performance. *Priority: High*

5.2 Safety Requirements

- **REQ-NF-008**: Display disclaimer on first launch and in settings: *"3AM Talks is not a substitute for professional mental health care. In emergency, contact [emergency services] immediately."* User must acknowledge before proceeding. Legal counsel and professional approval. *Priority: Critical*
- **REQ-NF-009**: Never provide medical diagnoses, prescribe medications, or recommend discontinuing treatment. LLM prompts explicitly prohibit diagnostic language. Content filtering blocks diagnostic terminology. Zero diagnosis instances in 500 test conversations. *Priority: Critical*
- **REQ-NF-010**: On system failure/error/connectivity loss, display emergency contacts and graceful error messages. No silent failures, comprehensive error handling. *Priority:*

High

- **REQ-NF-011**: Session timeout: 10-minute inactivity, 60-minute maximum (extendable). Timeout warnings 2 minutes prior. Sessions saved before timeout (with consent). *Priority: Medium*

5.3 Security Requirements

- **REQ-NF-012**: All client-server communication encrypted via TLS 1.2+ (TLS 1.3 recommended) with strong cipher suites (AES-256-GCM preferred, minimum AES-128). Perfect Forward Secrecy enabled. Certificate pinning on mobile apps. *Priority: Critical*
- **REQ-NF-013**: "Anonymous Mode": engage without PII, random UUID session IDs, no email/phone/name required, no device fingerprinting, auto-delete after 30 days inactivity. *Priority: High*
- **REQ-NF-014**: All data at rest encrypted (AES-256): database-level (PostgreSQL TDE), file system (mobile), key management via AWS KMS/Azure Key Vault, 90-day key rotation. *Priority: Critical*
- **REQ-NF-015**: Admin authentication: MFA required, RBAC (Admin, Developer, Auditor), audit logging, least privilege, quarterly access reviews. *Priority: High*
- **REQ-NF-016**: Rate limiting: 100 messages/hour per user, 500 requests/hour per IP. Exponential backoff for violations. 429 response with retry-after. *Priority: Medium*
- **REQ-NF-017**: Regular security audits: weekly automated vulnerability scanning, annual penetration testing, third-party audit before launch, critical vulnerabilities patched within 24 hours. *Priority: Medium*

5.4 Software Quality Attributes

5.4.1 Availability

99% target over 30-day periods (max 7.2 hours downtime/month). Monitoring via Pingdom/DataDog.

5.4.2 Reliability

- **REQ-NF-018**: Graceful handling of invalid/malformed inputs without crashes. Comprehensive input validation, error handling for empty/long/non-UTF-8 inputs. Fuzz testing shows no crashes after 10,000 random inputs. *Priority: High*

5.4.3 Usability

- **REQ-NF-019**: Initiate conversation within 2 taps from home screen. Average time to first conversation <30s including onboarding. >80% task completion rate for first-time users. *Priority: High*
- **REQ-NF-020**: System Usability Scale (SUS) score ≥75 (grade B) based on 30+ test users. *Priority: Medium*

5.4.4 Maintainability

- **REQ-NF-021**: Modular microservices architecture with clear service boundaries, OpenAPI-documented contracts, independent deployment, >80% unit test coverage. *Priority: Medium*

5.4.5 Portability

- **REQ-NF-022**: Deployable on AWS/Google Cloud/Azure without major changes. Infrastructure-as-Code (Terraform, Kubernetes), containerized services (Docker), cloud-agnostic abstraction layers. *Priority: Low*

5.4.6 Accessibility

- **REQ-NF-023**: WCAG 2.1 Level AA compliance: screen reader compatibility (iOS VoiceOver, Android TalkBack), color contrast ratios ≥4.5:1, keyboard-navigable elements, alternative text for non-text content. *Priority: High*

5.5 Business Rules

- **BR-001**: Only Administrators update therapeutic knowledge base (ensure clinical validity).
- **BR-002**: System provided free to end users (increase accessibility).
- **BR-003**: User data not sold, shared for marketing, or used for training external models without explicit consent (privacy protection).
- **BR-004**: GDPR compliance (European users): data processing agreements, privacy impact assessment, data protection officer, user rights (access, rectification, erasure) implemented.
- **BR-005**: Anonymized, aggregated usage data collected for research (ethics board approval): PII removed, group-level aggregation (≥10 users), re-identification risk reviewed.

6 UML Diagrams and System Architecture

6.1 Use Case View Diagram

The Use Case View illustrates how end users, administrators, crisis operators, and external AI services interact with the 3AM Talks system. It highlights core functionalities such as initiating therapy sessions, providing audio/video input, receiving empathetic responses, reviewing personality insights, and managing privacy settings. Administrators maintain system performance, update knowledge bases, and validate emotional models.

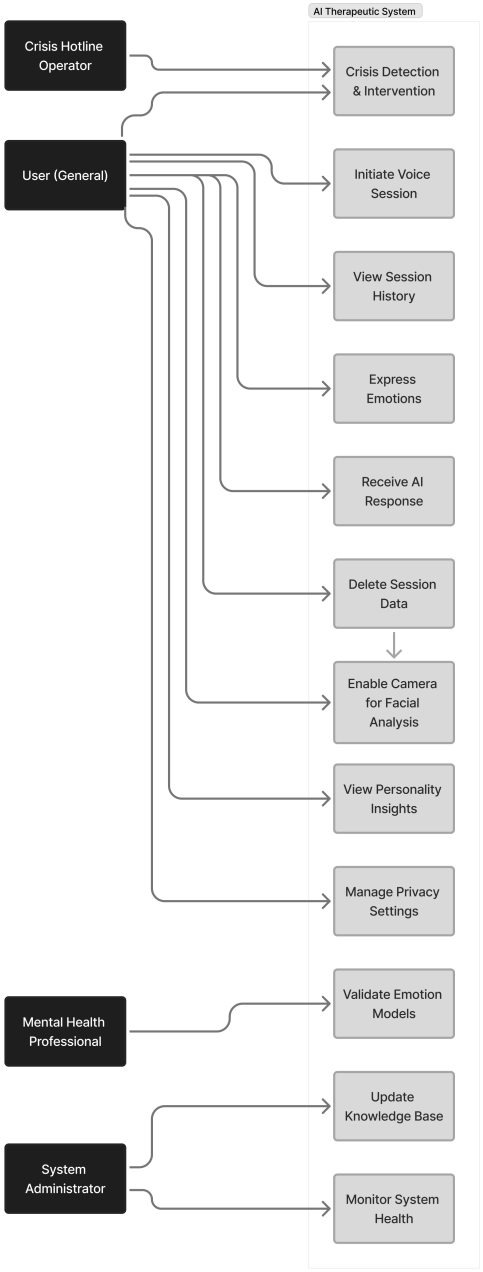


Figure 1: Use Case View Diagram for the 3AM Talks System

6.2 Data Flow Diagram

The Data Flow Diagram presents how multimodal data moves through the system. User audio and video streams are processed in parallel: Whisper performs transcription (P1), MediaPipe/OpenFace detects facial emotions (P2), and Hume AI analyzes vocal emotions (P3). NLP modules extract textual emotion cues (P4). The Emotion Fusion engine (P5) combines these signals, while the Personality Inference model (P6) updates user

trait profiles. The RAG engine retrieves relevant therapeutic content (P7), and the LLM generates responses (P8) which are delivered through Hume EVI-4 TTS (P9) and synchronized avatar animation (P10). Crisis detection modules (P11) continuously monitor content.

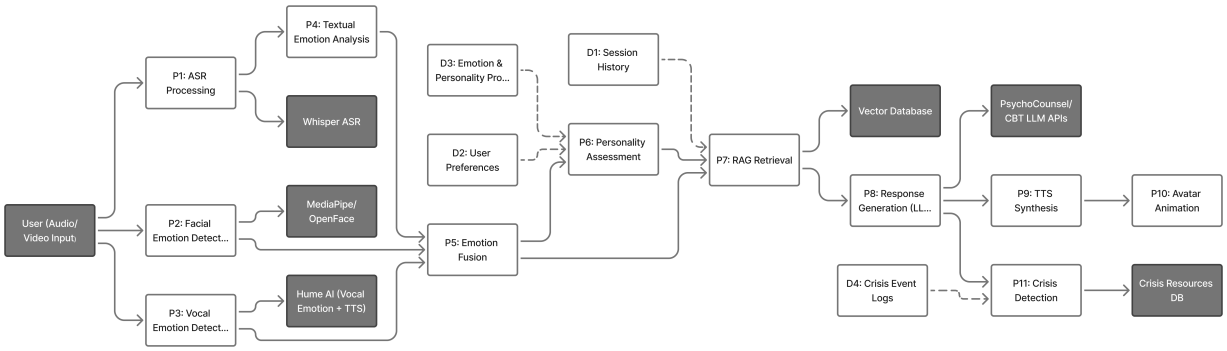


Figure 2: Data Flow Diagram for the 3AM Talks System

6.3 Sequence Diagram

The Sequence Diagram shows the chronological execution of the multimodal pipeline. When the user provides audio and video input, the system initiates parallel emotional processing:

- **U → M:** User sends audio/video stream.
- **M → FE:** Facial expression analysis begins.
- **M → VE:** Vocal emotion analysis begins.
- **M → TE:** Text emotion analysis begins.
- **FE → M:** Facial emotion results returned.
- **VE → M:** Vocal emotion results returned.
- **TE → M:** Text emotion results returned.
- **M → EF:** Emotion fusion engine combines all signals.
- **M → P:** Personality inference is updated.

These fused emotional and personality indicators are passed to the LLM, which generates a therapeutic response. It is then synthesized into emotional speech and animated through the 3D avatar.

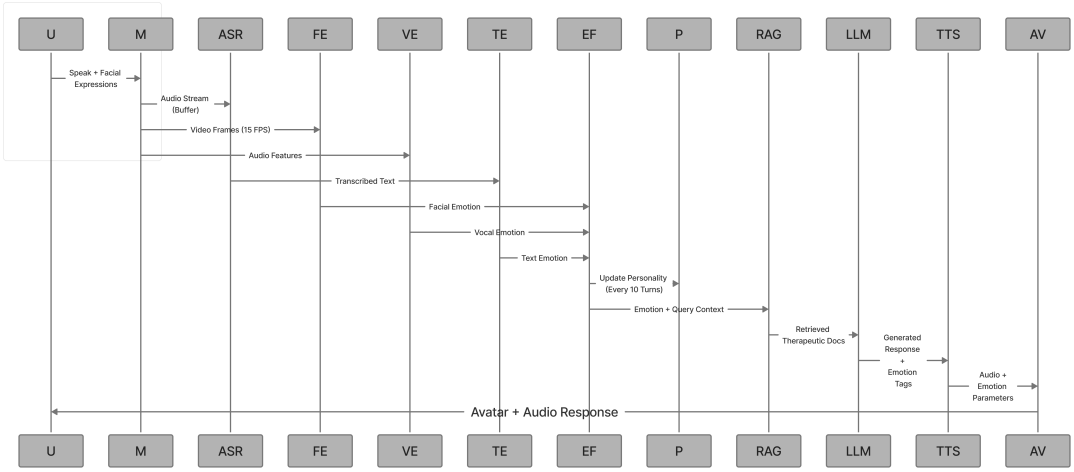


Figure 3: Sequence Diagram for the 3AM Talks System

6.4 Deployment Model

The Deployment Model illustrates how the system is distributed across user devices, network infrastructure, cloud microservices, AI processing units, and external service providers.

Client Layer: Mobile apps (iOS/Android) and web browsers capture audio/video, support WebRTC streaming, and render the WebGL-based 3D avatar. Local encrypted storage and offline emergency resources ensure safety.

Network Layer: All communication uses HTTPS/WSS with TLS 1.3. A CDN delivers static avatar assets, while global routing reduces latency.

Application Layer: FastAPI microservices run inside Docker containers on Kubernetes clusters. Components include ASR, emotion analysis services, RAG retrieval, personality modeling, LLM response generation, and avatar synthesis.

Data Layer: Encrypted PostgreSQL databases store session logs and preferences, vector databases support semantic retrieval, and Redis caches frequently accessed session data.

External Services: Includes Hume AI for vocal emotion and TTS, MediaPipe/OpenFace for facial emotion, and PsychoCounsel-Llama3-8B for therapeutic LLM responses.

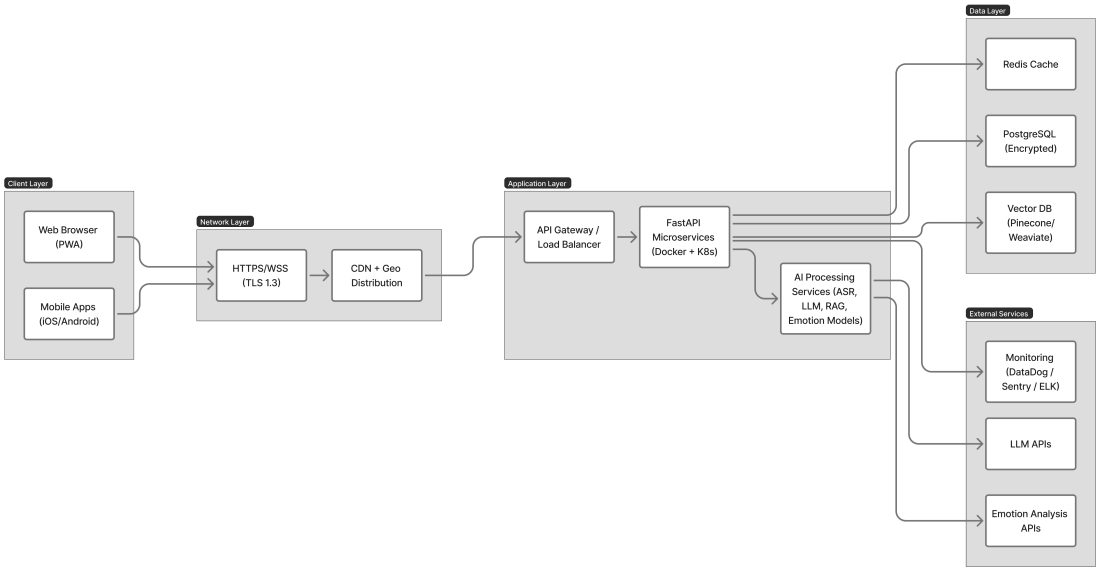


Figure 4: Deployment Model for the 3AM Talks System

A Glossary

Term	Definition
ACT	Acceptance and Commitment Therapy
API	Application Programming Interface
ASR	Automatic Speech Recognition (Speech-to-Text)
AES-256	Advanced Encryption Standard with 256-bit key
Barge-in	User interruption of system speech enabling natural flow
Big Five	Personality model: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism
CBT	Cognitive Behavioral Therapy
CDN	Content Delivery Network
DBT	Dialectical Behavior Therapy
Embedding	Dense vector representation in high-dimensional space
FPS	Frames Per Second (rendering performance metric)
GDPR	General Data Protection Regulation
LLM	Large Language Model
MFCC	Mel-Frequency Cepstral Coefficients (audio features)
MOS	Mean Opinion Score (speech quality metric, 1-5 scale)
NLP	Natural Language Processing
PII	Personally Identifiable Information
Prosody	Speech rhythm, stress, and intonation patterns
RAG	Retrieval-Augmented Generation
TLS	Transport Layer Security
TTS	Text-to-Speech
VAD	Voice Activity Detection
Vector Database	Specialized DB for high-dimensional similarity search
WCAG	Web Content Accessibility Guidelines
WebGL	Web Graphics Library (browser-based 3D rendering)
WER	Word Error Rate (ASR accuracy metric)
WSS	WebSocket Secure

B Analysis Models

B.1 Entity-Relationship Diagram (ERD)

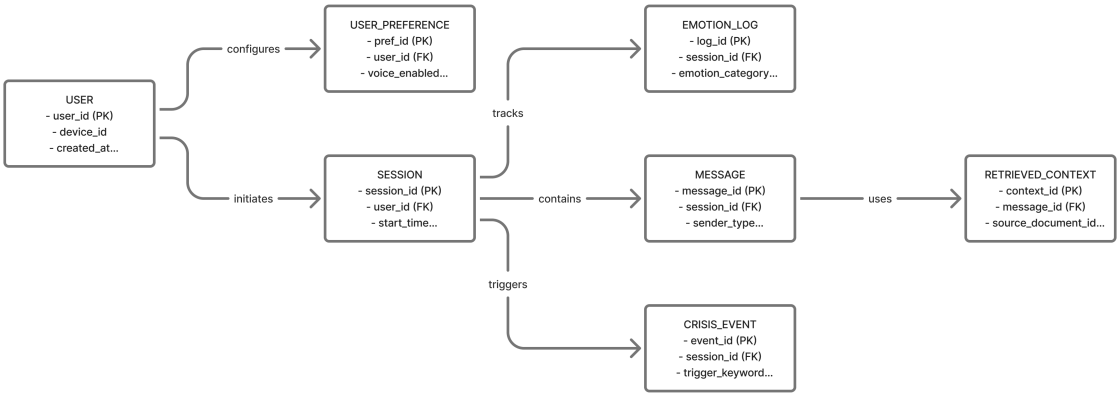


Figure 5: ERD for the 3AM Talks System

B.2 State Transition Diagram (Voice Interaction with Avatar)

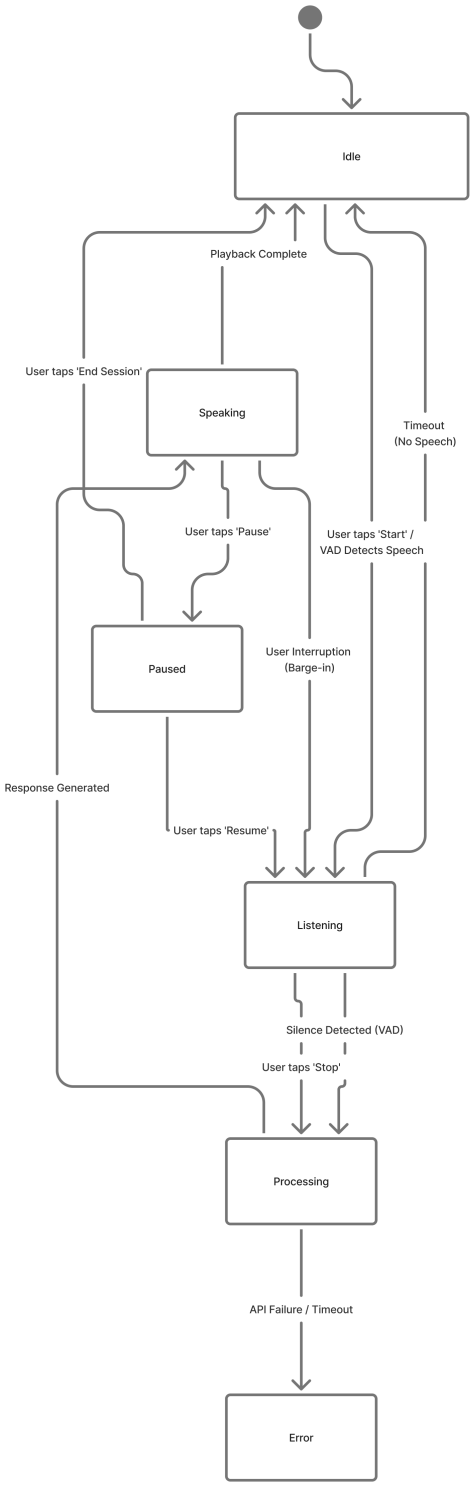


Figure 6: State Transition for the 3AM Talks System

C To Be Determined (TBD) List

ID	Section	Requirement	TBD Item	Target Resolution
TBD-01	2.5.3	Mental Health Regulations	Legal review of disclaimer	Alpha Release
TBD-02	4.2.3	Therapeutic Literature	Finalize 500+ docs	Beta Release
TBD-03	4.3.3	Emotion Fusion	Tune fusion weights	Beta Release
TBD-04	4.3.3	Avatar Design	Select 3D avatar models	Development Phase
TBD-05	4.5.3	Crisis Keywords	Validate keyword dictionary	Alpha Release
TBD-06	3.3.1	LLM Vendor	Decide self-host vs. API	Infrastructure Setup
TBD-07	3.3.2	Emotion API	Confirm facial emotion tool	Development Phase
TBD-08	5.3.3	Key Management	Select cloud KMS provider	Infrastructure Setup
TBD-09	2.2.2	Multilingual	Urdu model selection	Phase 2 Planning
TBD-10	5.2	Empathy Scoring	Develop scoring rubric	Beta Release
TBD-11	4.3.3	Personality Update Frequency	Optimize Big Five update frequency	Beta Release

End of Document

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